Pressure, Flow, and Blood Pressure: what to say

Built from the same content as the deck and the sheet, in the order the slides reveal it. Each paragraph is one click. The cues tell you what is on the screen; the words are a starting point for what you say, so change them into your own voice as you rehearse. Timing marks are rough, for a 75–minute lecture.

Opening. Last week the left ventricle ejected 70 mL into the aorta. Today we follow that blood around the circuit. We measure the pressure that pushes it, see how a tissue takes what it needs at the capillary, get the blood back up from the legs, and then look at how the body holds blood pressure steady over seconds and over days. One rule applies to all of it: flow equals the pressure difference divided by the resistance.

SHEET On the sheet, the first page is the drawing with numbered pins and a label list. Tell them: write the name on the numbered line as each pin drops, then fill the card for that number (what it is, what it does, why it matters) on the card pages at the back. Then the steps, then the effect box. The QR squares open any picture full size on their phone.

A. The vessels, and the rules of flow

about 12 min in

YOU SHOULD BE ABLE TO **Vessel structure and function.** Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance. **Pressure flow and resistance.** Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow.

SAY FIRST We start with the pipes. In the drawing, the thickness of each wall tells you what that vessel is built to do.

Label the drawing: One circuit, with the pressure in each vessel

CONTEXT BOX One lap of the systemic circuit, left to right: artery, arteriole, capillaries, venule, vein. The graph underneath is the pressure in each vessel. Wall thickness is drawn to tell you what each vessel is built to do. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the elastic artery. Aorta and its big branches: thick wall with layers of elastin. Stretches during ejection and recoils during diastole, so flow continues between beats. Why it matters: pressure reservoir: stiff arteries (age, atherosclerosis) raise systolic pressure and pulse pressure.

PIN 2 This is the arteriole. Small artery with a thick ring of smooth muscle and little elastin. The main site of resistance; its radius sets how much flow a tissue gets and how much pressure is left for the capillaries. Why it matters: total peripheral resistance lives here; this is where blood pressure drugs act.

PIN 3 This is the precapillary sphincter. A ring of smooth muscle where a capillary leaves a metarteriole. Opens or closes individual capillary beds. Why it matters: lets a tissue route blood to the cells that need it.

PIN 4 This is the capillary. One layer of endothelium, 5 to 10 µm across, with a basement membrane. The only vessel that exchanges: gases, nutrients, waste and fluid cross here. Why it matters: huge total cross-section makes flow slowest here, which gives time for exchange.

PIN 5 This is the venule. Small vein that collects capillaries. Some exchange continues; white cells leave the blood here in inflammation. Why it matters: first place pressure is low enough for cells to crawl out.

PIN 6 This is the vein. Thin wall, little muscle, large lumen, very compliant. Capacitance vessel: holds about 60% of the blood at rest and gives it up when the sympathetic system tightens it. Why it matters: the venous reservoir is what venous return and preload are drawn from.

PIN 7 This is the venous valve. Flaps of endothelium in limb veins. Let blood flow toward the heart only. Why it matters: make the skeletal muscle pump work; failed valves give varicose veins.

PIN 8 This is the endothelium. The single cell layer lining every vessel. Makes nitric oxide and endothelin, controls what crosses, keeps blood from clotting on it. Why it matters: endothelial damage is the start of atherosclerosis.

POP-UP DRAWING Click the drawing. The lining is not just a lining; it signals to the muscle under it. Blood dragging across it, and chemicals such as acetylcholine, histamine and bradykinin, switch on the enzyme that makes nitric oxide. Nitric oxide diffuses into the smooth muscle, raises cGMP, and the muscle relaxes. The same cells can make endothelin when they are injured, when flow is low, or under angiotensin II, and endothelin works through IP_3 and calcium to constrict. So the vessel makes its own dilator and its own constrictor. Nitroglycerin and sildenafil both work on the nitric oxide side.

PIN 9 This is the pulse pressure in the artery. The swing between systolic and diastolic in the arteries. Shows the stroke volume hitting an elastic wall. Why it matters: the pulse you feel at the wrist is this pressure wave, not the blood itself.

PIN 10 This is the pressure drop across the arterioles. The steepest fall on the profile, from about 93 to about 35 mm Hg. Resistance converts pressure into a controlled trickle for the capillaries. Why it matters: dilate the arterioles and pressure downstream rises; constrict them and it falls.

PIN 11 This is the capillary pressure. About 32 mm Hg at the arterial end falling to about 15 at the venous end. Pushes fluid out through the pores (filtration). Why it matters: this is the pushing force in the Starling balance.

PIN 12 This is the venous pressure. Low and steady, a few mm Hg, nearly zero at the right atrium. The small gradient that still drives blood back; it needs help from the pumps. Why it matters: central venous pressure is a bedside measure of volume status.

How it happens: One circuit, with the pressure in each vessel

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 The ventricle ejects into an elastic artery. During systole the aorta's elastic wall stretches and stores some of the pulse's energy; pressure peaks near 120 mm Hg.

STEP 2 Diastole: the artery recoils. The stretched wall springs back and keeps pushing blood forward between beats, so flow never stops and pressure only falls to about 80; the pulses on the graph are this stretch and recoil.

STEP 3 The arteriole resists. A thick ring of smooth muscle makes the arteriole the narrowest adjustable point; most of the pressure is spent getting through it (about 93 down to 35) and the pulse disappears.

STEP 4 The capillary exchanges. Total cross-sectional area is enormous, so velocity is slowest here; with one cell of wall, gases, nutrients, waste and fluid cross.

STEP 5 Venules and veins collect and store. Pressure is now a few mm Hg; the thin, compliant veins hold about 60% of the blood, and valves keep it moving one way.

STEP 6 Back to the right atrium at about 0 mm Hg. The whole loop runs on one rule: flow = pressure gradient ÷ resistance, so cardiac output = MAP ÷ total peripheral resistance.

EFFECT The effect, what changes because of this: Pressure falls all the way around, and most of it is spent at the arterioles. Resistance goes with $1 \div \text{radius}^4$, so halving an arteriole's radius raises its resistance 16 times: small changes in arteriolar tone are the body's main way to steer flow and set blood pressure. Give them a minute to write the effect box in their own words before you move on.

Fill in: The rules of flow

LINE 1 Flow = pressure gradient ÷ resistance ($F = \Delta P \div R$); for the whole circuit, CO = MAP ÷ TPR.

LINE 2 Resistance depends on vessel length, blood viscosity, and radius; R is proportional to length × viscosity ÷ radius^4.

LINE 3 Halve the radius and resistance rises 16 times, so flow falls to one sixteenth at the same pressure.

LINE 4 Double the viscosity (polycythemia) and resistance doubles, flow halves.

LINE 5 Length matters when new vessels grow (obesity, a tumor); otherwise it is fixed.

LINE 6 Velocity of flow is flow ÷ total cross-sectional area: fastest in the aorta, slowest in the capillaries, faster again in the veins.

Clicker to close section A

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. An arteriole's radius falls to half. With the same pressure gradient, what happens to flow through it?

A. Falls to half

B. Falls to one quarter

C. Falls to one sixteenth

D. Unchanged, because the heart compensates

Key: C. Resistance goes with $1 \div r^4$; half the radius is 16 times the resistance, so one sixteenth the flow.

Not A: That would be true if resistance went with $1 \div r$.

Not B: $1 \div r^2$ would give a quarter; the fourth power is what Poiseuille's relationship says.

Not D: The question is about this one vessel at a fixed gradient.

Set B. Which vessel is built to be a reservoir, and what feature makes it one?

A. Elastic artery; elastin recoils

B. Arteriole; thick smooth muscle

C. Vein; thin compliant wall and large lumen

D. Capillary; one cell thick

Key: C. Veins stretch easily, so they hold about 60% of the blood at low pressure and release it when the sympathetic system tightens them.

Not A: The elastic artery is a pressure reservoir, not a volume reservoir.

Not B: Arterioles are the resistance vessels.

Not D: Capillaries exchange; they hold little blood.

Set C. Blood velocity is slowest in the capillaries even though they are the narrowest vessels. Why?

A. Capillary blood is more viscous

B. Their total cross-sectional area is the largest, and velocity = flow ÷ area

C. Precapillary sphincters stop the flow

D. Pressure is highest there

Key: B. Flow is the same through every level of the circuit; spread over a huge total area it moves slowly, which gives time for exchange.

Not A: Viscosity does not change in the capillaries.

Not C: Sphincters route blood; they do not slow it across the whole bed.

Not D: Capillary pressure is low, about 32 mm Hg falling to 15, not high.

BRIDGE Keep the pressure graph in your head as we go: the big drop happens at the arterioles.

B. Arterial blood pressure, and how you measure it

about 36 min in

YOU SHOULD BE ABLE TO **Arterial blood pressure.** Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance. **Blood pressure measurement.**

Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge.

SAY FIRST You will take blood pressures in lab, so it is worth knowing exactly what the cuff is doing to the artery.

Label the drawing: The arterial pressure wave and the cuff

CONTEXT BOX Left: the pressure in a large artery, beat by beat. Right: the same wave with a cuff pressure (gold) being let down across it. The sounds you hear through the stethoscope are drawn below the wave. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the systolic pressure. The peak, about 120 mm Hg, during ventricular ejection. Set by stroke volume and how stiff the arteries are. Why it matters: this is the top number of a blood pressure reading.

PIN 2 This is the diastolic pressure. The trough, about 80 mm Hg, just before the next ejection. Set mainly by total peripheral resistance and how long diastole lasts. Why it matters: this is the bottom number of a blood pressure reading, and it tells you how tight the arterioles are.

PIN 3 This is the dicrotic notch. The small blip on the falling side of each pulse. Marks the aortic valve closing (S2) and the brief backflow against it. Why it matters: separates ejection from diastole on a pressure trace.

PIN 4 This is the pulse pressure. Systolic minus diastolic, about 40 mm Hg. Rises with a bigger stroke volume or stiffer arteries. Why it matters: a wide pulse pressure in an older adult means stiff arteries.

PIN 5 This is the mean arterial pressure. The average pressure over the cycle, about 93 mm Hg; diastolic + one third of pulse pressure. The pressure that actually drives flow to the organs. Why it matters: mAP = cardiac output × total peripheral resistance; the number the baroreflex defends.

PIN 6 This is the cuff pressure. The pressure in the inflated cuff, let down slowly. Above systolic it flattens the artery; below diastolic it does nothing. Why it matters: the sounds you hear only exist while cuff pressure is between the two.

PIN 7 This is the first Korotkoff sound. The first tapping heard as the cuff is let down. Cuff pressure has just dropped below systolic: blood spurts through the squeezed artery in turbulent bursts. Why it matters: the cuff reading at that moment is the systolic pressure.

PIN 8 This is the last Korotkoff sound. The sounds muffle and then stop. Cuff pressure has dropped below diastolic: the artery stays open and flow is smooth and silent. Why it matters: the cuff reading at that moment is the diastolic pressure.

How it happens: The arterial pressure wave and the cuff

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Cuff above systolic: the artery is shut. At 140 mm Hg the cuff squashes the brachial artery flat; no blood passes, no sound.

STEP 2 Cuff drops below systolic: the first tap. The first time the pulse peak exceeds cuff pressure, blood squirts through the narrowed artery in a turbulent burst; that tap is the first Korotkoff sound and the cuff reading is systolic.

STEP 3 Sounds continue while the cuff is between the two. Each beat the artery opens briefly and snaps shut: tapping with every pulse.

STEP 4 Cuff drops below diastolic: silence. The artery stays open all the time, flow is smooth (laminar) and makes no noise; the cuff reading when the sounds vanish is diastolic.

STEP 5 Pulse pressure and mean pressure. Pulse pressure = systolic − diastolic (120 − 80 = 40). Mean arterial pressure = diastolic + ⅓ pulse pressure (80 + 13 = 93), because diastole lasts longer than systole.

STEP 6 What sets each number. Systolic rises with stroke volume and stiff arteries; diastolic rises with resistance (arteriolar tone) and heart rate; MAP = cardiac output × total peripheral resistance.

EFFECT The effect, what changes because of this: Two numbers from a cuff tell you about the pump and the pipes: a high diastolic means the arterioles are tight, a wide pulse pressure means stiff arteries or a big stroke volume. MAP is the number the body defends, and it has only two levers: cardiac output and resistance. Give them a minute to write the effect box in their own words before you move on.

Fill in: Work the numbers

LINE 1 Pulse pressure = systolic – diastolic 120 – 80 = 40 mm Hg.

LINE 2 MAP = diastolic + ⅓ pulse pressure 80 + 40 ÷ 3 = 93 mm Hg (closer to diastolic because diastole lasts longer).

LINE 3 MAP = cardiac output × total peripheral resistance raise either and MAP rises; the baroreflex adjusts both.

LINE 4 A patient at 150/70 pulse pressure 80, MAP 97: stiff arteries (wide pulse pressure), MAP about normal.

LINE 5 A patient at 100/85 pulse pressure 15, MAP 90: small stroke volume with tight arterioles, as in early shock.

LINE 6 Posture or exercise in lab standing: a brief dip then recovery; exercise: systolic up, diastolic about the same or slightly down.

Clicker to close section B

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A cuff reading is 130/70. What is the pulse pressure and the mean arterial pressure?

A. 60 and 100

B. 60 and 90

C. 200 and 100

D. 130 and 70

Key: B. Pulse pressure 130 – 70 = 60; MAP = 70 + 60 ÷ 3 = 90.

Not A: 100 is the midpoint; MAP sits closer to diastolic because diastole is longer.

Not C: Pulse pressure is the difference, not the sum.

Not D: Those are the readings themselves.

Set B. While deflating a cuff you hear the first tap at 118 and the sounds disappear at 76. What was happening in the artery at 118?

A. The artery opened fully for the first time and flow became smooth

B. The pulse peak first exceeded cuff pressure, so blood spurted through in a turbulent burst

C. The cuff pressure dropped below diastolic

D. The artery closed for the first time

Key: B. The first Korotkoff sound is turbulent flow through a partly squeezed artery when cuff pressure just falls below systolic.

Not A: That is the moment the sounds stop, at diastolic.

Not C: Cuff pressure drops below diastolic at 76, not at 118.

Not D: The artery was already closed above systolic.

Set C. A patient's diastolic pressure rises from 80 to 100 with little change in systolic. Which variable most likely changed?

A. Stroke volume

B. Arterial stiffness

C. Total peripheral resistance

D. Heart rate fell

Key: C. Diastolic pressure is what the arterioles are doing; tight arterioles keep pressure from falling as far between beats.

Not A: Stroke volume mostly moves systolic and pulse pressure.

Not B: Stiff arteries raise systolic and widen pulse pressure.

Not D: A slower heart gives longer diastole and a lower diastolic pressure.

BRIDGE So a blood pressure gives you two numbers, one mostly about the pump and one mostly about the pipes, and the mean pressure between them is the one the body defends.

C. How an arteriole decides its own tone

about 70 min in

YOU SHOULD BE ABLE TO **Local control of blood flow.** Explain myogenic, metabolic, and endothelial control of arteriolar tone and predict the local flow response to increased tissue metabolism.

SAY FIRST Who decides how open an arteriole is? There are five answers, and the tissue the arteriole serves has most of the say.

Mini mind map: How an arteriole decides its own tone

CENTER Center of the map: Arteriolar tone. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 Myogenic. Stretch opens cation channels in the smooth muscle, it depolarizes and contracts; drop the pressure and it relaxes. Keeps flow steady when arterial pressure changes (autoregulation).

BRANCH 2 Metabolic (active hyperemia). A working tissue makes more CO_2, H^+, K^+ and adenosine and uses up O_2; all of them relax the arteriole. Flow follows metabolism: exercising muscle, a digesting gut, an active brain region.

BRANCH 3 Endothelial. The lining makes nitric oxide (cGMP, relax) in response to shear stress, ACh, histamine and bradykinin, and endothelin (contract) in response to injury. Nitroglycerin and sildenafil act on the NO side.

BRANCH 4 Reactive hyperemia. Block the flow, metabolites build up and the arterioles dilate; release the block and flow overshoots before settling. The same metabolic control, seen after an interruption.

BRANCH 5 Sympathetic tone. Tonic norepinephrine on α_1 receptors keeps most arterioles partly constricted; more firing constricts, less firing dilates. Local signals can override it: exercising muscle dilates despite high sympathetic activity.

Clicker to close section C

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. During a run, the arterioles in the leg muscles dilate even though sympathetic activity is high. What explains the dilation?

A. Parasympathetic nerves to the muscle arterioles

B. Local metabolites (CO_2, H^+, K^+, adenosine, low O_2) override sympathetic tone

C. The myogenic response to low pressure

D. Endothelin release

Key: B. Active hyperemia: the working tissue's own chemistry relaxes its arterioles, and local control wins over the sympathetic signal in that bed.

Not A: Skeletal muscle arterioles have essentially no parasympathetic supply.

Not C: Arterial pressure is up, not down, in exercise.

Not D: Endothelin is a constrictor, so it would narrow the arterioles, not dilate them.

Set B. Arterial pressure rises suddenly, yet flow to the kidney barely changes. Which mechanism did that?

A. Active hyperemia

B. The myogenic response: stretch made the arterioles constrict

C. Sympathetic withdrawal

D. Nitric oxide release

Key: B. Stretch opens cation channels in arteriolar smooth muscle, the cell depolarizes and contracts, and resistance rises to match the pressure: autoregulation.

Not A: No change in metabolism here.

Not C: That would dilate and raise flow.

Not D: NO would dilate and raise flow.

Set C. Nitroglycerin relieves chest pain by giving the vessel nitric oxide. Which pathway does it switch on?

A. $\alpha_1 \rightarrow IP_3 \rightarrow Ca^{2+}$ up → contract

B. Stretch → cation channels → depolarize

C. NO → cGMP → Ca^{2+} down and phosphatase on → relax

D. Endothelin → contract

Key: C. Nitric oxide from the endothelium (or a drug) raises cGMP in the smooth muscle, which lowers Ca^{2+} and activates myosin phosphatase: relaxation.

Not A: That is the constrictor pathway.

Not B: That describes the myogenic response, which constricts; it is not what dilates muscle arterioles in exercise.

Not D: Endothelin is the endothelium's constrictor.

BRIDGE Local signals win inside the tissue that needs the blood; sympathetic tone decides everywhere else.

D. Capillary exchange, and edema

about 116 min in

YOU SHOULD BE ABLE TO **Capillary exchange and Starling forces.** Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends. **Edema.** Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism. **Lymphatic return.** State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.

SAY FIRST Now the exchange itself, which is the reason the whole circuit exists. We need two forces, one capillary, and the lymphatic vessel that cleans up afterward.

Label the drawing: Starling forces along one capillary

CONTEXT BOX One capillary opened out, blood flowing left to right. Red arrows: pressure pushing fluid out through the pores. Blue arrows: the pull of plasma proteins drawing fluid back in. Green: the lymphatic that collects what is left over. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the capillary hydrostatic pressure, arterial end. Blood pressure inside the capillary at its start, about 32 mm Hg. Pushes fluid out through the pores. Why it matters: the force that wins at the arterial end.

PIN 2 This is the capillary hydrostatic pressure, venous end. Blood pressure at the capillary's end, about 15 mm Hg. Still pushes out, but weakly. Why it matters: at this end the osmotic pull is stronger than the push, so fluid comes back in.

PIN 3 This is the plasma colloid osmotic pressure. About 25 mm Hg of pull from plasma proteins, mostly albumin, that cannot leave the capillary. Draws water back in along the whole length. Why it matters: low albumin (liver failure, kidney loss, starvation) means fluid stays out: edema.

PIN 4 This is the interstitial fluid. The fluid between the cells, about a quarter of the extracellular fluid. Its own hydrostatic pressure is near zero and its protein is low, so its forces are small. Why it matters: edema is simply too much interstitial fluid.

PIN 5 This is the net filtration (arterial end). Out minus in: about 32 – 25 = 7 mm Hg outward. Fluid leaves the capillary carrying dissolved nutrients. Why it matters: filtration is how plasma, with its dissolved nutrients, reaches the cells.

PIN 6 This is the net absorption (venous end). In minus out: about 25 – 15 = 10 mm Hg inward. Most of the filtered fluid returns here. Why it matters: across the body about 3 L a day is left over, not returned.

PIN 7 This is the lymphatic capillary. Blind-ended vessel with overlapping flaps that open inward. Picks up the leftover 3 L a day and any protein that leaked, and returns it to the veins. Why it matters: block it and the limb swells (lymphedema).

PIN 8 This is the capillary wall (endothelium with pores). One cell thick, with pores between the cells (and fenestrations in kidney and gut). Water and small solutes pass; proteins mostly do not. Why it matters: inflammation widens the pores, proteins leak, and the osmotic pull collapses.

How it happens: Starling forces along one capillary

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 At the arterial end, hydrostatic pressure wins. Capillary hydrostatic pressure is about 32 mm Hg pushing out; plasma colloid osmotic pressure is about 25 pulling in; net 7 mm Hg outward: filtration.

STEP 2 Fluid leaves, proteins stay. Water and small solutes pass through the pores between endothelial cells; albumin is too big, so the osmotic pull inside stays at 25.

STEP 3 Pressure falls along the capillary. Friction drops hydrostatic pressure from 32 to about 15 by the venous end; the osmotic pull does not change.

STEP 4 At the venous end, the osmotic pull wins. 25 in against 15 out: net 10 mm Hg inward: absorption. Most of what left comes back.

STEP 5 The lymphatic takes the remainder. About 3 L a day across the body is not reabsorbed; lymphatic capillaries pick it up, with any protein that leaked, and return it to the veins near the heart.

STEP 6 The interstitial forces are small. Interstitial hydrostatic pressure is near zero and interstitial protein is low, so in health the balance is set by the two capillary forces.

EFFECT The effect, what changes because of this: Filtration at the start, absorption at the end, lymph for the difference. Raise capillary pressure, lower plasma protein, open the pores, or block the lymphatics and fluid piles up in the tissue: that is edema, and each cause maps to a disease. Give them a minute to write the effect box in their own words before you move on.

Mini mind map: Four ways to get edema

CENTER Center of the map: Edema. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 Raised capillary pressure. More hydrostatic push along the whole capillary: more filtration, less absorption. Right heart failure (ankles), left heart failure (lungs), a blocked vein, standing all day.

BRANCH 2 Low plasma protein. Less albumin means less osmotic pull back in; filtration continues, absorption fails. Liver failure, nephrotic syndrome, protein malnutrition (kwashiorkor).

BRANCH 3 Leaky capillaries. Inflammation and histamine widen the pores; albumin escapes and takes its osmotic pull outside with it. A swollen ankle, hives, burns, anaphylaxis.

BRANCH 4 Blocked lymphatics. The 3 L a day that is not reabsorbed has nowhere to go. Filariasis (elephantiasis), lymph nodes removed in breast cancer surgery.

Steps in order: Lymph, back to the blood: about 3 L a day

STEP 1 Interstitial space. Of the roughly 20 L filtered out of capillaries each day, about 17 L is reabsorbed at the venous end; the 3 L left behind, plus any protein that leaked, is lymph.

STEP 2 Lymphatic capillary. Blind-ended tube whose endothelial cells overlap like one-way flaps: fluid pushes them open to get in and shuts them to get out.

STEP 3 Lymphatic vessels. Valves and the skeletal muscle pump move the lymph along, the same way they move venous blood.

STEP 4 Lymph nodes. Filtered through nodes on the way: immune cells inspect it.

STEP 5 Thoracic duct, or right lymphatic duct. The thoracic duct collects from the whole body below the diaphragm and the left upper quarter; the right lymphatic duct takes the right upper quarter.

STEP 6 Subclavian veins. Both ducts empty into the junction of the internal jugular and subclavian veins, so the 3 L is back in the blood, and the plasma protein with it.

Clicker to close section D

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. At the venous end of a capillary, hydrostatic pressure is 15 mm Hg and plasma colloid osmotic pressure is 25 mm Hg. Which way does fluid move, and how strongly?

A. Out, 10 mm Hg

B. In, 10 mm Hg

C. Out, 40 mm Hg

D. No net movement

Key: B. 25 in against 15 out is a net 10 mm Hg inward: absorption.

Not A: This option has the direction backwards.

Not C: The forces subtract, they do not add.

Not D: They are not equal here.

Set B. A patient with liver failure has swollen ankles and ascites. Which Starling force is abnormal?

A. Capillary hydrostatic pressure is up

B. Plasma colloid osmotic pressure is down because albumin is low

C. Interstitial hydrostatic pressure is up

D. Capillary permeability is down

Key: B. The liver makes albumin; without it the pull back into the capillary is weak, so filtered fluid stays in the tissue.

Not A: That is the heart failure or venous obstruction mechanism.

Not C: Interstitial pressure rises as a result, it is not the cause.

Not D: Lower permeability would reduce edema.

Set C. After lymph nodes are removed from one armpit, that arm slowly swells. Why?

A. Capillary pressure in the arm rose

B. The arm's lymphatics can no longer return the 3 L a day of leftover filtrate and leaked protein

C. Albumin fell in that arm

D. The veins of the arm are blocked

Key: B. Filtration always exceeds absorption a little; without lymphatic drainage the difference accumulates: lymphedema.

Not A: Capillary pressure is unchanged by node removal.

Not C: Plasma albumin is the same everywhere.

Not D: The veins are intact; venous obstruction would raise capillary pressure instead.

BRIDGE Those are the four ways to get edema, and each one is a disease you will meet in clinical work.

E. Venous return

about 170 min in

YOU SHOULD BE ABLE TO **Venous return.** Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume.

SAY FIRST Blood has to get back up from the feet to the heart with almost no pressure left to push it. The body uses three pumps.

Label the drawing: Three pumps that bring blood back

CONTEXT BOX Left: the trunk with the diaphragm between thorax and abdomen and the vena cava running up through both. Middle: a leg vein with its valves and the calf muscle around it. Right: a vein before and after sympathetic constriction. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the venous valve. Flaps of endothelium in limb veins. Let blood flow toward the heart only. Why it matters: make the skeletal muscle pump work; failed valves give varicose veins.

PIN 2 This is the skeletal muscle pump. Leg muscles squeezing the deep veins each time they contract. Pushes blood toward the heart; the valves stop it falling back. Why it matters: standing still for a long time loses this pump and blood pools.

PIN 3 This is the respiratory pump: thoracic pressure falls on inspiration. The chest cavity's pressure drops as the ribs lift and the diaphragm descends. Draws blood up the vena cava toward the right atrium. Why it matters: deep breathing raises venous return, which is one reason it helps when you feel faint.

PIN 4 This is the respiratory pump: abdominal pressure rises. The descending diaphragm squeezes the abdomen. Pushes blood out of the abdominal veins toward the chest. Why it matters: the two halves of the respiratory pump work together.

PIN 5 This is the sympathetic venoconstriction. Norepinephrine on α_1 receptors in venous smooth muscle. Makes the veins less compliant, so the same volume sits at higher pressure and moves toward the heart. Why it matters: this is how the body pulls blood out of the venous reservoir in hemorrhage and in exercise.

PIN 6 This is the venous pooling in the legs. Blood settling in the compliant leg veins under gravity. Lowers venous return, end-diastolic volume and stroke volume. Why it matters: why soldiers standing at attention faint, and why you feel faint standing up fast.

PIN 7 This is the vena cava. The two great veins returning blood to the right atrium. Carry the venous return, at a pressure of only a few mm Hg. Why it matters: easily compressed; late pregnancy lying flat can squeeze it.

How it happens: Three pumps that bring blood back

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Standing: gravity fills the leg veins. Veins are compliant, so blood pools below the heart; venous return, end-diastolic volume and stroke volume all fall.

STEP 2 The skeletal muscle pump. Each contraction squeezes the deep veins; the valve above opens and the valve below closes, so blood can only go up.

STEP 3 The respiratory pump, inhale. Thoracic pressure falls as the chest expands, which draws blood up the vena cava toward the right atrium.

STEP 4 The respiratory pump, below the diaphragm. The descending diaphragm raises abdominal pressure and pushes abdominal venous blood toward the chest.

STEP 5 Sympathetic venoconstriction. Norepinephrine on α_1 receptors tightens the veins; the reservoir holds less, so blood moves to the heart.

STEP 6 More venous return, more stroke volume. Venous return fills the ventricle; by Frank-Starling the next beat ejects more.

EFFECT The effect, what changes because of this: Venous return is what preload is made of. Stand still for a long time and the pumps stop working: venous return, stroke volume and blood pressure fall and you can faint. Walk, breathe deeply, or let the sympathetic system tighten the veins, and the blood comes back. Give them a minute to write the effect box in their own words before you move on.

Clicker to close section E

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A guard stands motionless for an hour and faints. Trace the cause.

A. Sympathetic venoconstriction raised venous return too much

B. Blood pooled in the leg veins, venous return and stroke volume fell, MAP fell, brain flow fell

C. The respiratory pump pushed blood away from the heart

D. The baroreflex constricted the carotid arteries

Key: B. Without the skeletal muscle pump, gravity fills the compliant leg veins; less returns to the heart, so by Frank–Starling stroke volume falls, and pressure to the brain drops.

Not A: Venoconstriction raises return; it is the compensation, not the cause.

Not C: The respiratory pump helps return.

Not D: The baroreflex does not constrict cerebral arteries.

Set B. Why do deep breaths raise venous return?

A. Thoracic pressure rises and squeezes the vena cava

B. Thoracic pressure falls and abdominal pressure rises, pulling and pushing blood toward the chest

C. The lungs release blood

D. The diaphragm compresses the heart

Key: B. Inspiration lowers pressure in the chest and raises it in the abdomen; venous blood moves down the gradient toward the right atrium.

Not A: This option has the direction backwards.

Not C: The lungs do not release stored blood on inspiration.

Not D: The diaphragm descends, away from the heart.

Set C. A vein's valves have failed (varicose veins). Which pump stops working properly?

A. Respiratory pump

B. Skeletal muscle pump, because blood squeezed upward falls back down

C. Sympathetic venoconstriction

D. The heart

Key: B. The muscle pump needs valves: without them a contraction pushes blood both ways and the segment refills from above.

Not A: The respiratory pump acts on central veins.

Not C: Venoconstriction does not need valves.

Not D: The heart itself is not changed by failed leg valves; the problem is downstream in the veins.

BRIDGE Venous return is where preload comes from, so this is the supply side of the Frank-Starling relationship we met last week.

F. Holding pressure steady: the reflex and the hormones

about 236 min in

YOU SHOULD BE ABLE TO **Baroreceptor reflex.** Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure. **Hormonal control of blood pressure.** Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control. **Adrenal cortex.** Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each.

SAY FIRST Now the control systems. First the one that works in seconds, then the one that works over days.

Label the drawing: The baroreceptor reflex

CONTEXT BOX Left: the two places pressure is sensed, the carotid sinus and the aortic arch. Middle: the medulla. Right: the three targets. Dark lines carry the signal in; blue is the parasympathetic output, rust the sympathetic output. Read it for a fall in pressure, such as standing up. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the carotid sinus baroreceptor. Stretch receptors in the wall of the carotid sinus, where the common carotid splits. Fire faster when stretched by higher pressure, slower when pressure falls. Why it matters: guards pressure to the brain; pressing on the neck here can slow the heart.

POP-UP DRAWING Click the drawing. A baroreceptor is a stretch sensor, nothing more. At normal pressure the wall is stretched a little and the nerve endings fire a few action potentials a second. When pressure rises the wall stretches more and they fire faster. The message to the medulla is the firing rate, carried by the glossopharyngeal nerve from the carotid sinus and the vagus from the aortic arch. When pressure falls the rate drops, and the medulla reads that too. That is the whole sensor: more stretch, more spikes.

PIN 2 This is the aortic arch baroreceptor. Stretch receptors in the wall of the aortic arch. Same signal as the carotid sinus, for the whole body. Why it matters: both sets report continuously; the reflex is tonic, not just emergency.

PIN 3 This is the glossopharyngeal nerve (CN IX), afferent. The nerve carrying carotid sinus signals to the medulla. It carries sensory signals in; it is the afferent arm of the reflex for the carotid sinus. Why it matters: cut it and the medulla no longer hears from the carotid sinus.

PIN 4 This is the vagus nerve (CN X), afferent. Carries aortic arch signals to the medulla. Sensory input (the vagus also carries the parasympathetic output back out). Why it matters: the same nerve carries the sensory signal in and the parasympathetic signal back out.

PIN 5 This is the medullary cardiovascular control center. The nucleus of the solitary tract and the neighboring centers in the medulla. Compares baroreceptor input with the set point and adjusts the two autonomic outputs. Why it matters: this is the integrating center of the reflex, where the comparison with the set point happens.

PIN 6 This is the parasympathetic output to the SA node. Vagal fibers releasing acetylcholine on M_2 receptors. Slows the pacemaker. Why it matters: this is the fast brake on heart rate, and resting vagal tone is why the resting rate is near 70 rather than 100.

POP-UP DRAWING Click the drawing; it is the one from last week. Acetylcholine on M_2 receptors works through Gi: it lowers cAMP, so the funny channels open less, and it opens potassium channels, so the cell sits lower. The slope flattens and the floor drops, and threshold comes later. This is the fast brake, and because the vagus is firing all the time at rest, your resting rate sits near seventy instead of the node's own hundred.

PIN 7 This is the sympathetic output to the heart. Sympathetic fibers releasing norepinephrine on β_1 receptors in the SA node and ventricles. Raises heart rate and contractility. Why it matters: raises cardiac output within seconds.

POP-UP DRAWING Click the drawing; it is from last week too. Norepinephrine on β_1 receptors works through Gs and raises cAMP. In the SA node the funny channels open more and the slope gets steeper, so heart rate rises. In the ventricles the same cAMP opens more L-type calcium channels and speeds the SR pumps, so each beat is stronger and shorter. Rate and contractility both go up, which is why cardiac output can rise within seconds.

PIN 8 This is the sympathetic output to arterioles. Norepinephrine on α_1 receptors in arteriolar smooth muscle. Constricts arterioles, raising total peripheral resistance. Why it matters: resistance is one of the two terms in MAP = CO × TPR, so this output moves blood pressure quickly.

POP-UP DRAWING Click the drawing; it is the chemical door from last week. Norepinephrine lands on an α_1 receptor, the receptor turns on Gq, Gq turns on phospholipase C, IP_3 is made, IP_3 opens the SR calcium channel, and the receptor-operated channel lets more calcium in. The arteriole constricts, resistance rises, and since mean pressure is cardiac output times resistance, blood pressure rises right away. This is the output the baroreflex leans on hardest.

PIN 9 This is the sympathetic output to veins. Norepinephrine on α_1 receptors in venous smooth muscle. Venoconstriction raises venous return and stroke volume. Why it matters: pulls blood out of the reservoir when pressure falls.

How it happens: The baroreceptor reflex

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Pressure falls, the baroreceptors stretch less. Carotid sinus and aortic arch receptors fire slower.

STEP 2 Fewer signals reach the medulla. Along the glossopharyngeal (carotid sinus) and vagus (aortic arch) nerves to the cardiovascular control center.

STEP 3 The medulla cuts the parasympathetic brake. Less vagal acetylcholine on M_2 receptors: the SA node speeds up.

STEP 4 and raises sympathetic drive to the heart. Norepinephrine on β_1: heart rate and contractility rise, so cardiac output rises.

STEP 5 Sympathetic drive constricts the arterioles. Norepinephrine on α_1: total peripheral resistance rises.

STEP 6 and the veins. Venoconstriction raises venous return and stroke volume.

EFFECT The effect, what changes because of this: Within a few heartbeats cardiac output and resistance both rise and MAP comes back toward its set point: MAP = CO × TPR, and the reflex pushes both. It works the other way for a rise in pressure. It is fast but it does not hold a grudge: in chronic hypertension it resets to the new pressure within a day or two, so the long-term work belongs to the kidney and its hormones. Give them a minute to write the effect box in their own words before you move on.

Label the drawing: The renin–angiotensin–aldosterone system, with ADH and ANP

CONTEXT BOX The long-term lever on blood pressure: a chain of hormones that ends in the kidney keeping salt and water. Follow the arrows from the kidney at the left, through the blood, to the targets on the right. ANP, in blue, pulls the other way. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the juxtaglomerular cells (renin). Modified smooth muscle cells in the wall of the afferent arteriole of the nephron. Release renin when renal pressure falls, when the macula densa sees less NaCl, or when sympathetic β_1 receptors are stimulated. Why it matters: the kidney's own pressure sensor starts the cascade.

POP-UP DRAWING Click the drawing. Three things make these cells release renin, and all three mean the same thing to the kidney: pressure or volume is low. First, the cells are stretch sensors in the afferent arteriole wall; less stretch means less pressure, and they release renin. Second, sympathetic nerves release norepinephrine onto their β_1 receptors. Third, the macula densa next door tastes the tubular fluid; less sodium chloride going past means less filtrate, so it signals the JG cells. Renin goes into the blood and the cascade starts.

PIN 2 This is the angiotensinogen (from the liver). A plasma protein always present in blood. Renin cuts it to angiotensin I. Why it matters: nothing happens without the enzyme; the substrate is just waiting.

PIN 3 This is the angiotensin I. A ten amino acid peptide, inactive. Nothing yet; it circulates until an enzyme converts it. Why it matters: it is a step with no action of its own, which is why the next enzyme matters so much.

PIN 4 This is the angiotensin-converting enzyme (ACE). An enzyme on the endothelium, especially in the lungs. Converts angiotensin I to angiotensin II. Why it matters: aCE inhibitors (the -pril drugs) block this step to lower blood pressure.

PIN 5 This is the angiotensin II. The active hormone of the whole system. Constricts arterioles, triggers aldosterone and ADH, drives thirst, and raises sympathetic activity. Why it matters: it raises pressure right away by raising resistance, and over the following days by raising blood volume.

POP-UP DRAWING Click the drawing. One hormone, four targets, one direction. On arterioles it works through AT_1, IP_3 and calcium to constrict, which raises resistance now. On the adrenal cortex it releases aldosterone, which will raise volume over days. On the posterior pituitary it releases ADH, and in the hypothalamus it drives thirst, both adding water. And it turns sympathetic activity up. Pressure up now by resistance, up later by volume.

PIN 6 This is the aldosterone (adrenal cortex). A steroid from the zona glomerulosa. Makes the distal nephron reabsorb Na^+ (and water follows) and secrete K^+. Why it matters: the slow lever: more volume, more pressure, over hours to days.

POP-UP DRAWING Click the drawing. Aldosterone is a steroid, so it walks into the distal nephron cell and acts on genes. The cell makes more sodium channels and more sodium-potassium pumps, so more sodium is reabsorbed, water follows it, and potassium is secreted in exchange. Because it works through new proteins it is slow: hours to days. This is the slow lever on blood volume.

PIN 7 This is the antidiuretic hormone (ADH, vasopressin). A peptide from the hypothalamus released by the posterior pituitary. Makes the collecting duct reabsorb water; at high levels constricts arterioles. Why it matters: released by angiotensin II, by high osmolarity, and by a large fall in pressure.

POP-UP DRAWING Click the drawing. ADH binds V_2 receptors on the collecting duct cell, raises cAMP, and the cell inserts aquaporin-2 water channels into its membrane. Water leaves the tubule and the urine becomes concentrated. At high levels it also binds V_1 receptors on arterioles and constricts them, which is why it has a second name, vasopressin. It is released by angiotensin II, by a rise in osmolarity, and by a large fall in pressure.

PIN 8 This is the thirst center. Hypothalamic neurons that angiotensin II and high osmolarity stimulate. They make you feel thirsty and drink. Why it matters: volume cannot be replaced by the kidney alone; intake has to rise.

PIN 9 This is the atrial natriuretic peptide (ANP). A peptide released by the atria when they are stretched by too much volume. Makes the kidney excrete Na^+ and water, dilates arterioles, and inhibits renin, aldosterone and ADH. Why it matters: it is the brake on the whole system, released when there is already too much volume.

POP-UP DRAWING Click the drawing. This is the brake. When the atria are stretched by too much volume they release ANP. ANP makes the kidney excrete sodium and water, dilates arterioles, and inhibits renin, aldosterone and ADH, so it undoes the whole cascade. Everything else we have drawn pushes pressure up; this one pushes it down, and it is released only when there is already too much.

How it happens: The renin–angiotensin–aldosterone system, with ADH and ANP

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 The kidney releases renin. Juxtaglomerular cells release renin when renal perfusion pressure falls, when the macula densa senses less NaCl, or when sympathetic nerves fire on β_1 receptors.

STEP 2 Renin cuts angiotensinogen. A liver protein always present in plasma becomes angiotensin I, which is inactive.

STEP 3 ACE makes angiotensin II. Angiotensin-converting enzyme on the lung endothelium converts I to II; this is the step the -pril drugs block.

STEP 4 Angiotensin II constricts arterioles. Resistance rises at once: a fast effect.

STEP 5 Angiotensin II triggers aldosterone. The zona glomerulosa releases aldosterone; the distal nephron reabsorbs Na^+ and water follows: more volume over hours to days.

STEP 6 Angiotensin II triggers ADH and thirst. The posterior pituitary releases ADH (water reabsorbed in the collecting duct) and the hypothalamus makes you drink.

STEP 7 ANP opposes it. When the atria are overstretched they release atrial natriuretic peptide: the kidney excretes Na^+ and water, arterioles dilate, and renin, aldosterone and ADH are all inhibited.

EFFECT The effect, what changes because of this: The baroreflex fixes pressure in seconds by changing resistance and output; this system fixes it over hours and days by changing blood volume. Most blood pressure drugs act here: ACE inhibitors, angiotensin receptor blockers, aldosterone antagonists, and diuretics that undo the volume. Give them a minute to write the effect box in their own words before you move on.

Mini mind map: Hormones that hold pressure over days

CENTER Center of the map: Blood volume. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 Angiotensin II. Constricts arterioles now; triggers aldosterone, ADH and thirst for later; raises sympathetic activity. Everything else in this map is triggered by it.

BRANCH 2 Aldosterone. Steroid from the zona glomerulosa: the distal nephron reabsorbs Na^+ (water follows) and secretes K^+. Blood volume rises over hours to days.

BRANCH 3 ADH (vasopressin). From the posterior pituitary: aquaporins in the collecting duct reabsorb water; at high levels constricts arterioles. Released by angiotensin II, high osmolarity, or a large pressure fall.

BRANCH 4 Atrial natriuretic peptide. From stretched atria: excrete Na^+ and water, dilate arterioles, inhibit renin, aldosterone and ADH. This is the brake that keeps the other three from overshooting.

Label the drawing: The adrenal cortex in section

PIN 1 This is the capsule. The connective tissue coat of the gland. Holds the gland together. Why it matters: the zona glomerulosa sits right under it, which is how the zone got its name as the outer layer.

PIN 2 This is the zona glomerulosa. Outer cortical zone, cells in small clusters. Makes aldosterone (a mineralocorticoid) under the control of angiotensin II and plasma K^+. Why it matters: because it controls salt, it controls blood volume, and so blood pressure.

PIN 3 This is the zona fasciculata. Middle and thickest zone, cells in straight columns. Makes cortisol (a glucocorticoid) under ACTH. Why it matters: stress, blood glucose, and the immune brake; too much is Cushing syndrome.

PIN 4 This is the zona reticularis. Inner cortical zone, cells in a net. Makes the adrenal androgens (DHEA) under ACTH. Why it matters: these androgens drive pubic and axillary hair at puberty and make a small contribution in adults.

PIN 5 This is the adrenal medulla. The core: chromaffin cells, a modified sympathetic ganglion. Releases epinephrine and norepinephrine into blood. Why it matters: this is the fight-or-flight hormone source from last week.

Fill the table: The adrenal cortex: zone, hormone, action

TABLE Across then down, one cell per click. Columns: Zona glomerulosa, Zona fasciculata, Zona reticularis.

ROW Hormone. Zona glomerulosa: aldosterone (mineralocorticoid). Zona fasciculata: cortisol (glucocorticoid). Zona reticularis: adrenal androgens (DHEA).

ROW Controlled by. Zona glomerulosa: angiotensin II and plasma K^+. Zona fasciculata: ACTH from the anterior pituitary. Zona reticularis: ACTH.

ROW Primary action. Zona glomerulosa: kidney keeps Na^+ and water, secretes K^+: blood volume and pressure. Zona fasciculata: raises blood glucose, breaks down protein and fat, suppresses inflammation and immunity: the stress hormone. Zona reticularis: weak androgens; pubic and axillary hair at puberty, libido in women.

ROW Too much. Zona glomerulosa: hyperaldosteronism: hypertension, low K^+. Zona fasciculata: Cushing syndrome: central obesity, high glucose, thin skin. Zona reticularis: virilization.

Clicker to close section F

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. Blood pressure falls when a patient stands. In the next few seconds, which combination does the baroreflex produce?

A. More vagal firing, less sympathetic firing

B. Less vagal firing, more sympathetic firing: HR up, contractility up, arterioles and veins constrict

C. More vagal firing, more sympathetic firing

D. No change until the kidney responds

Key: B. Fewer baroreceptor signals make the medulla release the brake and press the accelerator; both cardiac output and resistance rise.

Not A: That is the response to a rise in pressure.

Not C: The two outputs move in opposite directions.

Not D: The kidney is the slow system; the reflex is immediate.

Set B. An ACE inhibitor is started. Which step stops, and what two downstream effects fall?

A. Renin release; aldosterone and ANP fall

B. Angiotensin I to angiotensin II; arteriolar constriction and aldosterone fall

C. Angiotensinogen production; ADH and thirst fall

D. Aldosterone action; Na^+ excretion falls

Key: B. ACE converts I to II; without angiotensin II there is less vasoconstriction and less aldosterone (and less ADH and thirst).

Not A: Renin actually rises, since the feedback inhibition by angiotensin II is lost.

Not C: The liver keeps making angiotensinogen.

Not D: Aldosterone falls because its trigger is gone; Na^+ excretion rises.

Set C. A patient's atria are stretched by fluid overload. Which hormone rises, and what does it do?

A. Aldosterone; keeps Na^+

B. ADH; keeps water

C. Atrial natriuretic peptide; the kidney excretes Na^+ and water and renin is inhibited

D. Angiotensin II; constricts arterioles

Key: C. ANP is the volume brake: natriuresis, vasodilation, and inhibition of renin, aldosterone and ADH.

Not A: Aldosterone would make the overload worse and is inhibited here.

Not B: ADH is inhibited by ANP.

Not D: Angiotensin II is suppressed when renin is inhibited.

BRIDGE The reflex is fast but it resets to whatever pressure becomes normal; the kidney is slow but persistent. Most blood pressure drugs act on the slow system.

G. Two stress tests: exercise, and losing blood

about 311 min in

YOU SHOULD BE ABLE TO **Cardiovascular response to exercise.** Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise. **Compensation in hemorrhage and shock.** Sequence the compensatory responses to blood loss and explain the point at which compensation fails.

SAY FIRST Two stress tests tie all of this together: exercise, and losing blood.

Fill the table: Moderate exercise: what changes

TABLE Across then down, one cell per click. Columns: Change, Why.

ROW Heart rate. Change: up, to 150 or more. Why: parasympathetic withdrawn, then sympathetic drive.

ROW Stroke volume. Change: up. Why: venous return up (muscle and respiratory pumps, venoconstriction) and contractility up.

ROW Cardiac output. Change: up 4 to 5 times, to 20 to 25 L/min. Why: HR × SV, both up.

ROW Total peripheral resistance. Change: down. Why: active hyperemia dilates muscle arterioles, which outweighs constriction elsewhere.

ROW Mean arterial pressure. Change: up a little; systolic up, diastolic about the same. Why: CO up more than TPR falls.

ROW Regional flow. Change: muscle up 10 times, skin up (heat), heart up; gut and kidney down; brain unchanged. Why: local dilation where it is needed, sympathetic constriction where it is not.

Steps in order: Losing blood: compensation, then failure

STEP 1 Blood volume falls. Less venous return, less EDV, less stroke volume, cardiac output and MAP fall.

STEP 2 The baroreflex fires within seconds. Heart rate and contractility up (β_1), arterioles constrict in skin, gut and kidney (α_1), veins constrict: pale, cool skin and a fast weak pulse; flow is kept for brain and heart.

STEP 3 Capillaries pull fluid back in. Lower capillary pressure means absorption beats filtration: interstitial fluid refills the plasma over minutes to hours (autotransfusion).

STEP 4 The hormones take over. Renin, angiotensin II, aldosterone and ADH keep Na^+ and water; thirst rises; the kidney makes little urine.

STEP 5 The slow repairs. Erythropoietin restores red cells over days; the liver replaces plasma proteins.

STEP 6 Where compensation fails. Past roughly 30 to 40% loss, MAP can no longer be held: the heart and the vasomotor center themselves are underperfused, acidosis relaxes the arterioles, capillaries leak, and the fall in pressure feeds itself (progressive, then irreversible shock).

Clicker to close section G

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. During moderate exercise, total peripheral resistance falls yet mean arterial pressure rises slightly. How?

A. Cardiac output rises more than resistance falls

B. Resistance rises in muscle

C. Stroke volume falls

D. The baroreflex is switched off

Key: A. MAP = CO × TPR; CO can rise four or five times while TPR falls, so the product still goes up a little.

Not B: Muscle arterioles dilate; resistance falls there.

Not C: Stroke volume rises with venous return and contractility.

Not D: The reflex is reset upward, not off.

Set B. Two hours after losing a liter of blood, a patient's hematocrit has fallen even though no fluid has been given. Why?

A. Red cells were destroyed

B. Interstitial fluid moved into the capillaries because capillary pressure fell and absorption exceeded filtration

C. The kidney excreted red cells

D. Aldosterone diluted the blood

Key: B. With lower capillary hydrostatic pressure the osmotic pull wins along the whole capillary; tissue fluid refills the plasma and dilutes the remaining red cells (autotransfusion).

Not A: Red cells are not destroyed by hemorrhage.

Not C: The kidney does not excrete red cells.

Not D: Aldosterone retains salt and water over hours to days; the early dilution is the capillary shift.

Set C. Why does shock become irreversible past a certain point even if the lost blood is replaced?

A. The baroreceptors are destroyed

B. Underperfused heart and brain centers fail, acidosis relaxes arterioles and capillaries leak, so the fall in pressure feeds itself

C. The kidney stops making renin

D. The veins can no longer constrict

Key: B. Once the organs doing the compensating (heart, vasomotor center) are themselves starved, the compensation collapses and a positive feedback loop takes over.

Not A: The receptors survive; the effectors fail.

Not C: Renin is high in shock.

Not D: Venoconstriction continues until the vasomotor center fails.

Closing. To sum up: resistance happens in the arterioles, exchange happens in the capillaries, the reservoir is in the veins, the fast control is the baroreflex, and the slow control is the kidney and its hormones. Next week we look at what the blood itself is made of. Work through the study page once with the sheet before then.